

Whole-Number Division: Quotients and Remainders

Equal groups and arrays, remainders in context, estimating quotients, and checking with multiplication

How to work:

- Write every division answer in two parts: the **quotient** (how many full groups) and the **remainder** (what is left over).
- The remainder is always **smaller than the number you are dividing by**. If it is not, you can make one more group.
- Check any division by multiplying back: (number in each group) $\times$ (number of groups) + (remainder).
- In a story, read the question again before you write the final answer — sometimes the leftover means you need one more group, and sometimes it is just left over.

1. **Circle equal groups.** The counters below show 27.

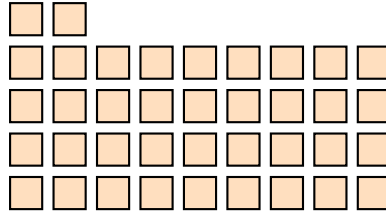


Circle groups of 4. How many full groups can you circle, and how many counters are left outside the circles?

quotient _____ remainder _____

Answer: _____

2. Build an array. The tiles below show 38.



Rearrange them into rows of 5. How many complete rows of 5 can you make, and how many tiles are left over?

quotient _____ remainder _____

Answer: _____

3. Check with multiplication. A baker has 45 rolls and puts 6 rolls in each bag. After filling the bags there are 3 rolls left on the tray.

$$6 \times \underline{\hspace{1cm}} + 3 = 45$$

How many bags did the baker fill? Show the multiplication that proves it.

Answer: _____

4. Estimate first. A club shares 632 prize tickets equally among 8 members.

- Round 632 to the nearest hundred to get a friendly number.
- Divide your rounded number by 8.

Write the rounded number and the estimated quotient.

Answer: _____

5. Vans for the field trip. 53 students are going on a field trip. Each van holds 8 students.

- Divide to find how many vans are completely full, and how many students are left without a seat.
- Then answer the question the trip organiser actually asked: how many vans must be booked?

Answer: _____ vans

6. **Share the marbles.** The marbles below show 74.



Nine friends share them equally. How many marbles does each friend get, and how many are left over? Draw a ring around one friend’s share.

Answer: _____

7. **Partial quotients.** A collector puts 145 stickers into an album, 6 stickers on each page. Fill in the table to find how many pages are filled and how many stickers are left.

Take away	Pages used	Stickers left
$6 \times 20 = 120$	20	

quotient _____ remainder _____

Answer: _____

8. Find the mistake. Nia divides 87 by 4 and writes

$$87 \div 4 = 20 \text{ R } 7$$

Use the remainder rule to explain how you can tell she is wrong without dividing again. Then write the correct quotient and remainder, and check them by multiplying back.

Answer: _____

9. Estimate, then divide. A library receives 1347 new books and puts 5 on each shelf.

- Round 1347 to the nearest hundred, then divide to estimate the number of shelves.
- Now divide exactly. Write the quotient and the remainder.

Answer: _____

10. Ribbon for bows. A ribbon is 96 centimetres long. Each bow uses 7 centimetres of ribbon.

- How many whole bows can be made, and how much ribbon is left over?
- The van problem needed one *extra* van for the leftover students, but this problem does not need one extra bow. Explain in your own words why the two leftovers are treated differently.

Answer: _____ cm